

HAZEM KHALED EZZAT

Data Analyst | Deep Learning and Computer Vision | Python, PyTorch, Power BI, SQL
Cairo, Egypt | +20 150 826 9570 | hazemelerefy@gmail.com
LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Data and Applied AI specialist with a focus on computer vision modeling, automated data pipelines, and production analytics. Combines deep learning research (PyTorch, YOLO) with enterprise SQL and Power BI architecture to build reliable, high-throughput data systems.

TECHNICAL SKILLS

- **Deep Learning and Computer Vision:** PyTorch, YOLOv11n, Computer Vision, Model Architecture Design, Feature Engineering, SHAP Analysis, Evaluation Metrics (mAP, AUC, F1)
- **Data Analytics and BI:** SQL (PostgreSQL), Python (Pandas, NumPy, Scikit-learn), Power BI, DAX, Excel, Data Warehousing, Relational Schemas (ERD), ETL Pipelines
- **Applied AI and Automation:** Autonomous Agents (n8n), Docker, FastAPI REST APIs, MCP Integration, Hugging Face Spaces, Linux, Git and GitHub
- **Front-End and UI:** React.js, Next.js, Three.js, TypeScript, JavaScript, Tailwind CSS, Figma
- **Development Tools:** VS Code, Cursor, Claude Code, GitHub Copilot, Jupyter Notebook, Google Colab

WORK EXPERIENCE

Data Analyst, Digilians (MCIT)

Cairo, Egypt | December 2025 to August 2026

- Built self-service Power BI KPI dashboards and DAX data models, automating recurring metrics and cutting ad-hoc reporting requests by 30%.
- Engineered automated SQL reporting workflows in PostgreSQL, replacing manual spreadsheet extraction and cutting report turnaround time by 50%.
- Designed and documented normalized relational database schemas with full ERD specifications to keep data consistent across reporting systems.
- Led technical research and architecture for DAFEsteeel, directing a 6-member team through computer vision modeling, Dockerized API containerization, and preprint submission.
- Built JobPulse, an automated job-market intelligence pipeline, and created automated AI content workflows using n8n and REST APIs.
- Communicated data trends and model findings to executive stakeholders through clear visual presentations and interactive dashboards.

Front-End Developer and UI Designer, Freelancer.com

Cairo, Egypt (Remote) | January 2021 to November 2025

- Built responsive web applications and client interfaces using HTML5, CSS3, JavaScript, React.js, and Next.js.
- Designed UI/UX wireframes, visual assets, and component systems in Figma with an emphasis on visual hierarchy and accessibility.
- Optimized front-end performance, asset delivery, and mobile responsiveness across client projects to maintain 95+ PageSpeed scores.

PROJECTS

DAFEsteel: Real-Time Deep Learning System for Industrial Steel Defect Detection

Lead Researcher and Architect | Graduation Project, Digilians (MCIT) | December 2025 to August 2026
[GitHub Repository](#) | [Live Demo on Hugging Face](#)

- Designed DAFEGate (Defect-Aware Feature Enhancement Gate), a lightweight plug-in module for YOLOv11n with dual branches: Sobel edge detection for linear cracks and texture-variance filtering for surface anomalies.
- Boosted mAP@0.5 from 79.35% to 81.98% on the NEU-DET benchmark (+5.5 percentage points on crazing specifically) while adding only 3.7% parameter overhead at 145 FPS real-time throughput.
- Ran 18 tracked ablation experiments validating architecture design choices, establishing that additive residual connections eliminated training instability found in multiplicative gates.
- Built the production deployment pipeline, including a Dockerized FastAPI REST service and a CLIP-based image filter to reject non-steel inputs.

NeuroScope: 3D Deep Learning Architecture Builder

Interactive Visual Workspace | React 18, Three.js, Zustand, Tailwind CSS, Vite
[GitHub Repository](#)

- Built an interactive 3D web application where users visually assemble neural network architectures and automatically export matching PyTorch or TensorFlow code in real time.
- Implemented a model zoo, custom 3D GLB components, command palette, and full keyboard navigation to configure optimizers, loss functions, and learning rates without mouse interaction.

Autonomous AI Agents and Intelligence Pipelines

Automation Systems | n8n, REST APIs, OpenAI API, Reddit API, Hacker News API
[GitHub Repository](#)

- Built an unsupervised Social Intelligence Pipeline in n8n running daily to ingest trending developer topics from five subreddits with automated rate-limiting, retries, and clean feed normalization.
- Created a multi-platform content publishing agent that scores trending articles from Hacker News and Dev.to with custom relevance algorithms and publishes Arabic summaries twice daily.

EDUCATION

Diploma in Applied AI and Data Analytics, Digilians (MCIT)

December 2025 to August 2026

- Hands-on curriculum in Python, SQL, machine learning, PyTorch deep learning, and AI agent automation. Graduation project: DAFEsteel.

Nanodegree in Front-End Web Development, Egypt FWD (Udacity)

September 2020 to March 2022

- Project-based program sponsored by Egypt's Ministry of Communications covering React.js, component architecture, and REST API integration.

Bachelor of Laws (LL.B.) in Commercial and Corporate Law, Port Said University

2020 to 2024

- Coursework in corporate governance, compliance, and contract analysis, providing structured analytical reasoning applied to data governance and enterprise data integrity.

CERTIFICATIONS

- **Microsoft Certified: Power BI Data Analyst Associate**, Microsoft (Credential ID: 2D0273674013E250, Earned July 2026)
- Google Data Analytics Professional Certificate, Google and Coursera
- Delivering Quality Work with Agility, IBM and Coursera

LANGUAGES

- **Arabic:** Native
- **English:** Professional Working Proficiency (B2)